

UCS664: Conversational AI: Natural Language Processing

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Course Objective: This course provides a broad introduction to deep learning and natural language processing. It offers some of the most cost-effective approaches to automated knowledge acquisition in the emerging field of natural language understanding using deep learning and GPU Computing.

Syllabus

Introduction: Natural Language Processing and its Applications, Introduction to Deep Learning, NVIDIA Toolkits, SDKs and platforms for Deployment.

Introduction to Natural Language Processing: What is NLP, Principles and Traditional Methods, Linguistics, Why Machine Learning and Deep Learning.
{Theory, Basics , Lectures}

Introduction to Deep Learning: Introduction to Artificial Neural Networks (ANNs), Artificial Neurons, Layers, Perceptron, Multilayer Perceptron, Advanced Deep Neural Networks (DNNs), Batch Normalization, Hyperparameter tuning, Activation Functions, Metrics, Optimization, Regularization.

Introduction to NLP using Deep Learning: Embeddings: Feedforward NN, Word2Vec, GloVe, Contextualization (ELMo etc.), Deep Recurrent Models: RNNs, GRUs, LSTMs.
{Lectures and DLI}

Advanced NLP using Deep Learning: Introduction to NeMo, Self-Attention, Transformer Networks: BERT and its Variants, Megatron etc, Working with open source datasets: GLUE Benchmarks.

Applications of NLP: Exploring NLP Problem Statements- Information Retrieval, Intent Slot Filling, Machine Translation, Punctuation & Capitalization, Question and Answering Machine, Relation Extraction, Sentiment Analysis, Token Classification in NeMo.
{General Papers, Blog , Survey Papers, Nemo}

Introduction to NVIDIA Toolkits and SDKs: Transfer Learning Toolkit, TensorRT Optimization, Triton Inference Server for Inferencing and Deployments, Various Visualization Tools, Kubernetes Deployment.

Laboratory Work:

1. Introduction to DL Frameworks: TLT, PyTorch, and Tensorflow (Keras).
2. Binary Classification with Perceptron and Logistic Regression.
{DLI Online Course: Getting Started with Deep Learning.}
{DLI Online Course: Deep Learning at Scale with Horovod.}
{DLI Online Course: Modeling Time-Series Data with Recurrent Neural Networks in Keras.}
3. Neural Modules (NeMo) for Training Conv AI Models, Exploring NeMo Fundamentals, Exploring NeMo Model Construction, Nemo Swap App Demo.
4. Sentiment Analysis & Text Classification with NeMo.
5. Intent Slot Filling for ChatBot using Joint Bert Model with NeMo.
6. Machine Translation with NeMo.
7. Question & Answering Machine with NeMo.
8. Information Retrieval, Punctuation & Capitalization, Relation Extraction, Sentiment Analysis, Token Classification with NeMo.

9. Hands-on practical on TensorRT Optimization, Triton Inference Server.

Course Learning Objectives (CLO)

After the completion of the course the student will be able to:

1. Understand the concept of Neural Networks and its implementation in the context of Machine Learning.
2. Exploring GPU computing for building deep learning based text analytics applications.
3. Deep understanding Conversational AI Toolkit: NeMo for training Deep Neural Network.
4. Practical Work around on Various NVIDIA SDKs for training and deployment: AMP, Horovod, TensorRT, Triton and so on.

Text Books

1. Schmidhuber, J. (2015). "Deep Learning in Neural Networks: An Overview". Neural Networks 61: 85-117.
2. Bengio, Y., LeCun, Y., and Hinton, G. (2015). "Deep Learning". Nature 521: 436-44.
3. Allen, James, Natural Language Understanding, Second Edition, Benjamin/Cumming, 1995.
4. Bengio, Y., Courville, A., and Vincent, P. (2013). "Representation learning: A review and new perspectives", IEEE Transactions on Pattern Analysis and Machine Intelligence 35 (8): 1798-1828.
5. Deep Natural Language Processing course offered at the University of Oxford:
<https://github.com/oxford-cs-deepnlp-2017/lectures>
6. "The Unreasonable Effectiveness of Recurrent Neural Networks" by Andrej Karpathy:
<https://karpathy.github.io/2015/05/21/rnn-effectiveness/>
7. Manning, Christopher and Heinrich, Schütze, Foundations of Statistical Natural Language Processing, MIT Press, 1999.

Reference Books:

1. Arel, I., Rose, D.C., and Karnowski, T.P. (2010). "Deep Machine Learning - A New Frontier in Artificial Intelligence Research". IEEE.
2. Computational Intelligence Magazine 5 (4): 13-18. Bengio, Y. (2009) "Learning deep architectures for AI". Foundations and trends in Machine Learning 2 (1): 1-127.
3. Radford, Andrew et. al., Linguistics, An Introduction, Cambridge University Press, 1999.
4. Christopher D. Manning and Hinrich Schütze. 1999. Foundations of Statistical Natural Language Processing. MIT Press.
5. "Understanding LSTM Networks" by Christopher Olah:
<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>